

AI2200: CONCENTRATION INEQUALITIES

PRACTICE PROBLEMS

TOPICS: Markov's Inequality, Chebyshev's Inequality, Chernoff Bound

1. Suppose that $X_1, X_2, \dots \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p)$ for some fixed constant $p \in (0, 1)$. For a fixed $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^n X_i$.

(a) Derive the moment generating function of S_n .

(b) Argue that for any $\varepsilon > 1$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - p\right| \geq \varepsilon\right) = 0$$

(c) Prove that for any $\varepsilon \in (0, 1)$,

$$\mathbb{P}\left(\left\{\frac{S_n}{n} - p \geq \varepsilon\right\}\right) = \begin{cases} 0, & p + \varepsilon \geq 1, \\ \leq \exp(-n D_{\text{KL}}(\text{Ber}(p + \varepsilon) \parallel \text{Ber}(p))), & 0 < p + \varepsilon < 1, \end{cases}$$

where $D_{\text{KL}}(\text{Ber}(p + \varepsilon) \parallel \text{Ber}(p))$ denotes the Kullback-Leibler divergence between $\text{Ber}(p + \varepsilon)$ and $\text{Ber}(p)$ distributions, and is defined as

$$D_{\text{KL}}(\text{Ber}(p + \varepsilon) \parallel \text{Ber}(p)) := (p + \varepsilon) \log \frac{p + \varepsilon}{p} + (1 - (p + \varepsilon)) \log \frac{1 - (p + \varepsilon)}{1 - p}.$$

Hint:

Use Chernoff's bound to obtain an upper bound for $\mathbb{P}(\{S_n > n(p + \varepsilon)\})$, and optimise the upper bound.

(d) Prove that for any $\varepsilon \in (0, 1)$,

$$\mathbb{P}\left(\left\{\frac{S_n}{n} - p < -\varepsilon\right\}\right) = \begin{cases} 0, & p - \varepsilon \leq 0, \\ \leq \exp(-n D_{\text{KL}}((1 - p + \varepsilon) \parallel (1 - p))), & 0 < p - \varepsilon < 1. \end{cases}$$

Hint: Set $Y_n = 1 - X_n$ for each $n \in \mathbb{N}$, and use the result in part (c) on the sum of $Y_1, \dots, Y_n$.

2. Let X be a random variable. Suppose that there exists a positive constant $\sigma > 0$ such that the MGF of X satisfies

$$M_X(t) \leq \exp\left(\frac{t^2 \sigma^2}{2}\right) \quad \forall t \in \mathbb{R}.$$

Prove that for any $\varepsilon > 0$,

$$\mathbb{P}(\{|X| > \varepsilon\}) \leq 2 \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right).$$

3. In lecture 02, we saw that if X is Pareto distributed, i.e.,

$$f_X(x) = \frac{3}{x^4}, \quad x \geq 1,$$

then for any $\alpha \in \mathbb{R}$,

$$\mathbb{P}(\{X \leq \alpha\}) \leq \exp\left(\inf_{t < 0} \{\log M_X(t) - t\alpha\}\right).$$

Notice that the above bound is vacuous if the exponent is non-negative. In this exercise, we will identify the values of α for which the exponent is negative, hence making the above bound meaningful.

(a) Compute $\mathbb{E}[X]$.

(b) Using the fact that

$$\left. \frac{dM_X(t)}{dt} \right|_{t=0} = \mathbb{E}[X],$$

identify all values of α for which the exponent in the bound is negative.

Hint: For a fixed α , let

$$\psi_\alpha(t) := \log M_X(t) - t\alpha.$$

Determine the slope of ψ_α at $t = 0$. What can you say about the exponent in the bound if this slope is positive?

4. Consider $K \geq 1$ Gaussian distributions $\mathcal{N}(\mu_m, \sigma_m^2)$, $m = 1, \dots, K$. For $n \in \mathbb{N}$, let $A_1, \dots, A_n$ be IID samples from a *mixture of the K Gaussian distributions*: for each $i \in \{1, \dots, n\}$, the random variable A_i is drawn from $\mathcal{N}(\mu_m, \sigma_m^2)$ with probability w_m . Here, $(w_1, \dots, w_K)$ is a known probability vector, i.e., $\sum_{m=1}^K w_m = 1$.

Let $\theta > 0$ be a fixed *radius* parameter. The sample A_i is said to be *typical for cluster m* if it is drawn from $\mathcal{N}(\mu_m, \sigma_m^2)$ **and** satisfies $|A_i - \mu_m| \leq \theta \sigma_m$. For each $m \in \{1, \dots, K\}$, let

$$V_m := \{i \in \{1, \dots, n\} : A_i \text{ is typical for cluster } m\},$$

and for each $i \in \{1, \dots, n\}$ and $m \in \{1, \dots, K\}$, let

$$p_m := \mathbb{P}(\{i \in V_m\}) = \mathbb{P}(\{A_i \text{ is typical for cluster } m\}).$$

For all the questions below, we fix a cluster index $m \in \{1, \dots, K\}$.

- (a) Let $Y_i := \mathbf{1}\{i \in V_m\}$. Notice that the cardinality of V_m is $|V_m| = \sum_{i=1}^n Y_i$. Compute $\mathbb{E}[|V_m|]$.
 (b) Are $Y_1, \dots, Y_n$ independent? Justify clearly.
 (c) Using $\text{Var}(Y_i) = p_m(1 - p_m) \leq 1/4$, show that for any $\varepsilon > 0$,

$$\mathbb{P}(\{|V_m| \leq (p_m - \varepsilon)n\}) \leq \frac{1}{4\varepsilon^2 n}.$$

Hint: Use Chebyshev's inequality.

- (d) Show, using the Chernoff bounding technique, that for any $\varepsilon > 0$,

$$\mathbb{P}(\{|V_m| \leq (p_m - \varepsilon)n\}) \leq \exp(-2n\varepsilon^2).$$

- (e) Show that

$$p_m \geq w_m \left(1 - 2 \exp\left(-\frac{\theta^2}{2}\right)\right).$$